SHUBHANKAR TIWARI

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EDUCATION

Columbia University — New York, NY
M.S. in Artificial Intelligence. — Expected Dec 2027

SRM Institute of Science and Technology — Chennai, Tamil Nadu
B.Tech, GPA: 9.44 — Sep 2022

WORK EXPERIENCE

Bank of America — Hyderabad, India
Software Engineer I A — Aug 2024 - Jul 2026

- Owned OpenShift deployments, CI/CD, release validation, and production stabilization across 17+ deployments.
- Ran modernization initiatives across 13 backend services including JDK 17, Spring 6, containerization, and cloud enablement, improving security and modernization readiness. Orchestrated AI adoption initiatives for 50+ CashPro colleagues.
- Diagnosed and resolved 50+ production issues involving SSL certificates, database connectivity, deployment failures, Drools dependencies, and privacy defects. Received 3+ awards.
- Invented a Bank of America patent under filing (2026) for a governed multi-modal AI architecture for financial fraud. Led 10+ team discussions to refine the idea, architecture, and presentation.

Bank of America — Hyderabad, India
Software Engineer I B — Apr 2023 - Aug 2024

- Owned CashPro Search: deployed features, resolved defects, became SME, and received 4+ Global Recognition Awards.

Bank of America — Hyderabad, India
Apprentice — Apr 2022 - Apr 2023

- Developed CashPro Search features using Java, Spring Boot, and Agile practices.

HighRadius Technologies — Delhi, India
Product Development Trainee — Aug 2021 - Mar 2022

- Developed Java and Python services for Order-to-Cash and deduction management systems, delivering production changes.

PROJECTS

Open Source (GitHub) — Feb 2026 - Jul 2026
Drift-Aware Fraud Detection and Model Governance

- Built a drift-aware ML governance pipeline using PSI/KL-divergence detection and automated retrain-evaluate-promote control on 284K transactions.
- Maintained 0.975 AUC-ROC with business-loss-aware gates (10:1 FN/FP weighting) blocking unstable or higher-loss candidates.

Open Source (GitHub) — Sep 2025 - Dec 2025
AI Reliability Platform for LLM Evaluation

- Built an LLM reliability platform using entropy-aware routing, adaptive regeneration, and behavioral workflows across 213 prompts in 5 categories.
- Engineered an adversarial noise injector and async pipeline, achieving 100% harmful-output detection on 43 prompts across 9 holdout rounds on fine-tuned mT5.
- Deployed a 9-category guardrail classifier with escalation logic, recording 0% leakage and 0% hallucination across 213 evaluations.
- Designed 55+ regression tests and CI-gated promotion workflows, blocking 7/7 unsafe candidates and reducing behavioral fragility 66% (25.0% → 8.5%).

TECHNICAL SKILLS

AI/ML: Python, PyTorch, Transformers, Scikit-learn, XGBoost.
ML Systems: LLM Evaluation, AI Reliability, Model Governance, Behavioral Testing, Guardrails, CI/CD for ML.
Backend: Java, Spring Boot, REST APIs, SQL, OpenShift, Jenkins, Ansible | Kaggle: Notebooks Expert, #707 Personal Best, 10 Bronze Medals.